

RESEARCH PLAN

Decentralized Swarm Consensus and Remote Vital Sign Detection for Low-Cost VTOL UAVs in Communication-Constrained Disaster Response

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Category	Engineering Technology

I. Research Problem

1.1 Engineering Question

Can a distributed consensus algorithm enable three low-cost, 3D-printed VTOL drones to maintain effective disaster-zone search and triage under fifty percent packet loss, while each drone simultaneously measures survivor heart rate, respiratory rate, and body temperature from a hovering position using imaging photoplethysmography (iPPG) and a thermal camera, and automatically classifies survivor movement to compute a complete Rapid Aerial Triage Score (RATS)?

Secondary Questions

- What is the minimum communication bandwidth required to maintain swarm task-allocation efficiency above 85% of a perfect-communication baseline?
- How does iPPG heart-rate accuracy (mean absolute error, MAE, in bpm) degrade with hover instability, standoff distance (5–15 m), and lighting variation — and does motion-artifact rejection via optical flow compensation preserve MAE below 10 bpm across all conditions?
- Does the CHROM chrominance-based rPPG algorithm offer statistically significant accuracy improvement over the green-channel FFT approach under drone vibration, and if so, at what computational cost on the Jetson Nano?
- Can a simple sensor-fusion rule based on RATS produce triage priority that agrees with ground-truth chest-strap and capnograph data with Cohen's kappa > 0.80 , including in the clinically critical ranges (HR < 60 or > 100 bpm; RR < 12 or > 20 bpm)?

- Does a Gilbert-Elliott bursty channel model change gossip-protocol resilience compared with uniform random drop, and if so, what gossip fan-out parameter preserves 85% task-allocation efficiency?

1.2 Practical Need and Rationale

Disaster first responders face a compound crisis during the golden hour: no aerial system exists that simultaneously searches for survivors, measures vital signs remotely, and maintains coordination when communication infrastructure is destroyed. The consequences are concrete. Manual search-and-triage of a 200 m × 200 m debris field by a two-person team takes approximately 40–60 minutes. For a survivor in hemorrhagic shock or severe hypothermia, those minutes are lethal.

This project fills four simultaneous gaps in the literature and in operational practice:

- Aerial vital sign measurement: no drone has demonstrated heart-rate measurement from a hovering VTOL at close range. This project extends iPPG from stationary hospital cameras to a dynamic, vibration-affected airborne platform.
- Decentralized swarm under communication loss: existing systems (DARPA OFFSET, Skydio, Anduril) rely on centralized ground-station coordination that fails when repeaters are destroyed. A gossip-protocol consensus algorithm continues operating with no ground link.
- Cost and repairability: existing disaster drones cost \$15,000+ per unit and cannot be repaired in the field. R.A.M.P.A.T. costs under \$1,000 and can be rebuilt from \$2 of PLA filament.
- Automated triage scoring: no prior system has computed a multi-parameter triage score (RATS) autonomously from drone-acquired remote vital signs, eliminating the need for a trained medic to assess each survivor.

Estimated impact: Simulation-based analysis (Section 6) projects that a three-drone consensus swarm assesses all survivors in a 200 m × 200 m area with 10 casualties in approximately 8 minutes, compared with 34 minutes for random single-drone search — a 76% reduction. This estimate is derived from the simulation trial data (mean time-to-assess across 500 random-search trials vs. 500 consensus trials at 0% packet loss) and will be validated against field test results.

1.3 Solution Criteria (Quantitative Success Thresholds)

Metric	Threshold	Rationale
HR accuracy (MAE)	< 10 bpm vs. Polar H10	Clinical triage threshold spacing is 20 bpm (RATS bands)

RR accuracy (MAE)	< 3 brpm vs. Nonin WristOx2 / capnograph	RATS band spacing is 4–8 brpm; 3 brpm error is clinically safe
Triage kappa	> 0.80 vs. ground truth RATS	APA standard for strong agreement; validated in Red/Yellow/Green ranges
Swarm speed-up	> 2× vs. random search baseline	Doubles usable time within the golden hour
Comms resilience	> 85% task-allocation at 50% packet loss (bursty model)	Represents realistic RF in concrete rubble structures
Cost per drone	< \$1,000 USD	Below mass-casualty logistics budget threshold
Flight endurance	> 30 min per charge	Covers a standard triage sweep of a 200 m × 200 m field
Processing latency	< 1 s from frame capture to priority display	First responder decision cycle time
Movement classification accuracy	> 85% vs. human observer label	RATS movement category; required for complete autonomous scoring

1.4 Constraints

- Weight: < 2 kg per drone (FAA micro UAS category, no waiver required).
- Altitude: < 120 m AGL (FAA Part 107 ceiling).
- Communication range: > 500 m from ground station.
- iPPG standoff: 5–15 m (face must resolve to at least 40 × 40 px at the sensor).
- Environmental: no precipitation; wind < 6 m/s.
- GPS: standard GNSS required; GPS-denied operation is out of scope (acknowledged limitation; addressed in Discussion).
- Human subjects: UCF IRB exemption filed; all volunteers provide written informed consent; no identifiable data retained; data stored on encrypted drive.
- RF environment: field tests conducted in FAA-authorized open area; WiFi and telemetry bands cleared with site coordinator to minimize co-channel interference.

1.5 Scope Exclusions

This system provides vital signs and triage priority to first responders. It does not perform autonomous evacuation, see through solid rubble, make medical diagnoses, operate in GPS-denied environments (GNSS signal blocked by heavy reinforced concrete), or deploy in

real disaster conditions. These exclusions are documented explicitly so that performance claims are not extrapolated beyond the tested operational envelope.

II. Background and Prior Work

2.1 R.A.M.P.A.T. (Prior ISEF Project)

R.A.M.P.A.T. is a VTOL tiltrotor drone with 178-minute flight time, a custom four-layer STM32F405 PCB flight controller, 73% drag reduction validated by CFD simulation, and a provisional patent filed (Application No. on file). Its limitation is single-agent operation with no swarm coordination and no vital sign measurement. The present project adds those capabilities as a direct extension.

2.2 State of the Art and Literature Gap

UAV swarm search: DARPA OFFSET and academic implementations use centralized communication architectures. No published system combines low-cost VTOL hardware with decentralized consensus under realistic packet loss. Wang et al. (2020) provide the theoretical basis for directed-graph consensus topologies for multiple UAVs but include no medical sensing.

Remote photoplethysmography (rPPG): Poh et al. (2010) demonstrated non-contact cardiac pulse measurement from stationary cameras using blind source separation. De Haan and Jeanne (2013) introduced the CHROM algorithm, which separates the PPG signal in chrominance space and is substantially more robust to motion artifact than single-channel methods. Both algorithms run on stationary platforms. No published work demonstrates rPPG from a hovering VTOL drone at 5–15 m.

UAV-based triage: Mösch et al. (2024) demonstrated iPPG from a tilt-wing drone achieving 89% triage accuracy on 15 subjects under gusty wind. That study used a single drone with centralized communication and did not implement a validated multi-parameter triage score. The present project extends this to a three-drone decentralized swarm with a complete RATS pipeline.

RATS score: RATS (Rapid Aerial Triage Score) is adapted from the National Early Warning Score 2 (NEWS2; Royal College of Physicians, 2017) by removing parameters that require physical contact (SpO_2 , blood pressure) and substituting remotely measurable proxies. Blood pressure is replaced by movement classification (proxy for perfusion state). RATS has not been previously published as a standalone instrument; this project constitutes its first experimental validation. The kappa criterion (> 0.80) is validated specifically in the Red and Yellow ranges where clinical decisions are most time-sensitive.

Communication models: Prior swarm simulation studies model packet loss as i.i.d. Bernoulli drops. Real RF propagation in disaster rubble structures follows a Gilbert-Elliott two-state Markov model (burst-error model) with average burst length 3–6 packets. This project is the first

UAV swarm study to validate consensus resilience under a bursty channel model in addition to the uniform random drop baseline.

Novelty summary: This project is the first to combine (1) iPPG heart-rate measurement from a hovering VTOL drone, (2) gossip-based consensus under bursty RF packet loss, (3) autonomous movement classification for complete RATS scoring, and (4) sub-\$1,000 field-repairable hardware. Each element individually has partial precedent; the combination does not.

III. Design and Methodology

3.1 Exploration of Alternatives

3.1.1 rPPG Algorithm Selection

Two candidate algorithms were evaluated for the iPPG pipeline prior to prototype implementation:

Algorithm	Principle	Vibration Robustness	Jetson Nano Load
Green-ch. FFT (GREEN)	FFT peak on green channel time series	Low: green channel captures motion artifact	< 5% CPU at 10 Hz
CHROM (de Haan 2013)	Chrominance-space projection cancels specular reflection	High: motion artifact projects onto luminance, not chrominance	~12% CPU at 10 Hz
POS (Wang 2017)	Plane orthogonal to skin color; adaptive to illumination	High: illumination-adaptive	~15% CPU at 10 Hz

Decision: CHROM is the primary algorithm. POS is implemented as a secondary comparison. GREEN is retained as the low-complexity baseline to quantify the accuracy gain from the more sophisticated approach. All three run in parallel during field tests; the primary RATS score uses CHROM output. This three-way comparison directly tests whether the additional computational cost of CHROM is justified on the drone platform.

3.1.2 Consensus Protocol Selection

Three distributed task-allocation strategies are compared:

- Random search: no communication; random waypoint navigation. Lower bound baseline.

- Centralized (perfect comms): all data relayed to ground station; ground station commands every drone. Upper bound baseline; not achievable in field deployment.
- Gossip consensus (proposed): each drone broadcasts to k random neighbors every 500 ms; $k = 2$ under normal conditions, $k = 3$ when packet loss exceeds 30% (adaptive fan-out).

Two channel models are tested for each protocol: (a) uniform i.i.d. Bernoulli drop and (b) Gilbert-Elliott two-state Markov burst model with mean burst length $L = 4$ packets (parameterized from empirical measurements of 2.4 GHz WiFi in a concrete-block obstacle course). This ensures the resilience claim generalizes beyond the idealized channel assumption used in prior literature.

3.1.3 Thermal Detection Approach

Two survivor detection methods are compared: (a) threshold-based blob detection (temperatures 32–40°C, contour area > 30 px at 10 m) and (b) a lightweight MobileNetV2 classifier fine-tuned on 200 FLIR Lepton thermal images of seated and prone adults. Method (a) is the primary implementation; method (b) is a secondary comparison to bound the accuracy gain from ML-based detection. If the classifier provides > 5% improvement in recall at a false-positive rate < 5%, it replaces threshold detection in the final pipeline.

IV. System Architecture

4.1 Hardware per Drone

Component	Specification
Flight controller	Custom ARM Cortex M7 PCB (4-layer); handles motor mixing, tilt-servo control, potentiometer feedback
Onboard compute	NVIDIA Jetson Nano Orin 8 GB; runs ROS2 Humble, iPPG pipeline, thermal detection, consensus algorithm
Frame	3D-printed PLA/PETG; < \$25 filament for full repair
Motors / ESCs	3 × 2212 1000 kV motors; 3 × 30 A BLHeli ESCs
Thermal camera	FLIR Lepton 3.5 (160×120 px, 8–14 μm , NETD < 50 mK); calibrated against Fluke 62 MAX IR thermometer pre-flight and mid-flight
RGB camera	Arducam IMX519 on 2-axis brushless gimbal; 60 fps, 4608×3456 native, cropped to 1920×1080 for iPPG
Range sensor	VL53L1X ToF (accuracy ± 20 mm at 4 m); used to maintain constant standoff for iPPG normalization

Obstacle avoidance	4 × HC-SR04 ultrasonic sensors (forward, left, right, rear)
Navigation	u-blox NEO-M8N GPS; MPU-6050 IMU; BMP388 barometer
Communication	ESP8266 WiFi (ROS2 swarm messages); 915 MHz SiK telemetry radio (ground station link, long range)
Power	4S 10000 mAh LiPo; INA219 current sensor for SOC estimation
Storage	MicroSD (all telemetry, video, consensus logs)

4.2 Clock Synchronization

Accurate timestamps are critical for the consensus conflict-resolution rule (keep the report with the higher confidence and the more recent timestamp). Without synchronization, clock drift between Jetson Nanos can cause the wrong report to win when two drones assess the same survivor within seconds of each other.

Implementation: All Jetson Nanos run chrony as an NTP client syncing to the ground-station laptop (stratum 2 server). The ground station syncs to pool.ntp.org when internet is available pre-flight, then holds its local clock. Expected inter-drone clock error: < 10 ms. This is verified at the start of each field trial by logging synchronized timestamps across all three drones and confirming max offset < 50 ms. If GPS PPS becomes available (NEO-M8N exposes the PPS pin), PPS-disciplined chrony (stratum 1) will reduce error to < 1 ms.

4.3 Thermal Camera Calibration

A single pre-flight blackbody calibration is insufficient for the FLIR Lepton because the sensor self-heats during flight (focal plane temperature rises ~3°C over 30 min), shifting the radiometric output by up to 1.5°C. This error straddles the 35°C and 39°C RATS thresholds, making accurate temperature scoring impossible without correction.

Calibration protocol:

- Pre-flight: 5-minute warm-up in shaded conditions; flat-field correction (FFC) triggered; blackbody reference (Fluke 62 MAX, accuracy $\pm 1^\circ\text{C}$) used to record a gain/offset calibration table at 34°C, 36°C, 38°C, and 40°C.
- Mid-flight (automated): The Jetson monitors the Lepton die temperature via I2C every 60 seconds. If die temperature exceeds the pre-flight baseline by > 2°C, it triggers an automatic FFC and applies a temperature-dependent correction offset derived from the pre-flight calibration table. FFC requires 200 ms of shutter closure; the drone briefly holds position during this interval.
- Post-flight validation: a subset of thermal frames from each trial is compared to Garmin Fenix 6 skin-temperature readings (accuracy $\pm 0.5^\circ\text{C}$) worn by each volunteer to compute residual error after correction.

Expected post-correction accuracy: $\pm 0.8^{\circ}\text{C}$ (1σ), sufficient to reliably distinguish the 35°C and 39°C RATS thresholds.

V. Signal Processing Pipelines

5.1 iPPG Heart-Rate Extraction (CHROM Primary)

Step	Implementation
1. Face detection	MediaPipe FaceMesh at 10 Hz; falls back to YOLOv8-nano (TensorRT) if MediaPipe confidence < 0.7
2. ROI definition	Forehead + bilateral cheek ROIs (each 30% of face width); cheeks excluded if partially occluded
3. Motion compensation	Two-level: (a) KCF tracker aligns ROI centroid frame-to-frame; (b) dense optical flow (Farnebäck, OpenCV) corrects sub-pixel residual drift. Linear trend subtracted from each channel.
4. CHROM projection	$X_s = 3R - 2G$; $Y_s = 1.5R + G - 1.5B$; $\alpha = \sigma(X_s)/\sigma(Y_s)$; $S = X_s - \alpha Y_s$ (skin-tone normalized pulse signal)
5. GREEN baseline	Parallel: bandpass green channel; FFT peak; retained for algorithm comparison
6. Bandpass filter	4th-order Butterworth; 0.7–3.3 Hz (42–198 bpm); slightly wider than original to tolerate tachycardia in trauma survivors
7. Spectral estimation	Welch PSD (4-second window, 50% overlap); dominant peak multiplied by 60 for HR in bpm
8. Confidence score	SNR of the dominant spectral peak vs. total power in band; confidence > 0.7 required for the reading to enter the RATS computation
9. Motion rejection gate	If estimated ROI displacement (from optical flow) exceeds 8 px between consecutive frames, that frame is excluded from the windowed average

5.2 Respiratory Rate Extraction

The respiratory pipeline targets the clinically actionable range of 6–30 brpm:

- Chest ROI: detected below the face landmark (sternum region, estimated from neck-to-hip proportion using MediaPipe pose keypoints).

- Vertical displacement: KCF tracker on chest ROI; vertical centroid displacement extracted across frames.
- Filter: 4th-order Butterworth bandpass, 0.1–0.5 Hz (6–30 brpm).
- FFT + Welch PSD: dominant peak frequency $\times 60 = \text{RR in brpm}$.
- Ground truth comparison: Nonin WristOx2 pulse oximeter (respiratory rate output via waveform, accuracy ± 1 brpm in 6–60 brpm range) on each volunteer. This replaces the Garmin watch which has a reported MAE of 2–3 brpm — too close to the 3 brpm target to serve as a reference standard.

5.3 Movement Classification (RATS Category)

The RATS movement category (Walking/Standing = 0 pts; Lying with limb movement = 1 pt; Lying still = 2 pts) requires an autonomous classification pipeline. This was identified as a gap in prior drone triage proposals and is addressed here explicitly.

Implementation:

- Input: thermal + RGB video frames at 10 Hz.
- Pose estimation: MediaPipe Pose (BlazePose GHUM Lite, TensorRT-optimized). Outputs 33 body keypoints with visibility scores.
- Classification rules: if no keypoints detected above torso = Lying still (score 2). If torso keypoints present but feet keypoints absent or near torso keypoints = Lying with movement (score 1). If full skeleton detected with vertical displacement of foot keypoints > 30 px across 3 consecutive frames = Walking/Standing (score 0).
- Confidence threshold: movement score is assigned only if at least 12 keypoints have visibility > 0.5 . Below this threshold, movement is treated as unknown and RATS movement contribution is set to 0 (conservative; underestimates severity slightly).
- Validation: a human observer independently labels movement category for each trial segment from the video recording; Cohen's kappa is computed between autonomous classification and human label. Target: kappa > 0.85 for movement classification alone.

5.4 RATS Triage Scoring

Parameter	Ranges and Points	Source
Heart Rate (bpm)	< 40 or $> 130 = 2$ pts $40\text{--}60$ or $100\text{--}130 = 1$ pt $60\text{--}100 = 0$ pts	CHROM iPPG
Respiratory Rate (brpm)	< 8 or $> 30 = 2$ pts $8\text{--}12$ or $21\text{--}30 = 1$ pt $12\text{--}20 = 0$ pts	Chest motion tracker
Temperature ($^{\circ}\text{C}$)	< 35 or $> 39 = 2$ pts $35\text{--}36$ or $38\text{--}39 = 1$ pt $36\text{--}38 = 0$ pts	Calibrated FLIR Lepton

Movement	Lying still = 2 pts Lying + movement = 1 pt Walking/Standing = 0 pts	MediaPipe Pose classifier
Total Score	0–2 = Green (minor) 3–4 = Yellow (delayed) 5–8 = Red (immediate)	Fusion

Ground truth RATS score is computed from: Polar H10 chest strap (HR, ± 1 bpm), Nonin WristOx2 (RR, ± 1 brpm), Garmin Fenix 6 skin temperature ($\pm 0.5^\circ\text{C}$), and human observer movement label. The drone-derived RATS score is compared to this ground-truth score using Cohen’s weighted kappa (linear weights) to account for the ordinal nature of Green/Yellow/Red categories.

5.5 Swarm Consensus Algorithm

5.5.1 Task Types and Utility Function

Each drone autonomously selects among five task types: SEARCH (explore unmapped area), ASSESS (hover and collect vitals), VERIFY (re-assess if HR change > 20 bpm), RELAY (forward messages for low-battery neighbor), RETURN (home if battery $< 15\%$).

Utility function for drone i and task j :

$$U(i, j) = w_1 \times \text{Priority}(j) + w_2 \times (1 - \text{dist}(i, j) / d_{\text{max}}) + w_3 \times \text{Battery}(i) / \text{Battery}_{\text{max}}$$

Weights are adaptive: early mission $w_1 = 0.3$, $w_2 = 0.5$, $w_3 = 0.2$ (prioritize coverage); after first survivor detected $w_1 = 0.6$, $w_2 = 0.25$, $w_3 = 0.15$ (prioritize triage). Weight transitions are triggered by the first survivor detection event broadcast to all drones.

5.5.2 Gossip Protocol with Adaptive Fan-out

Every 500 ms each drone broadcasts its survivor list (position, vitals, RATS score, confidence, NTP-synchronized timestamp) to k randomly selected neighbors. Adaptive fan-out: $k = 2$ when estimated packet loss $< 30\%$ (computed from ACK receipt rate over the last 10 broadcasts); $k = 3$ when loss $\geq 30\%$.

Conflict resolution: if two drones report the same survivor (GPS positions within 3 m), the Jetson retains the report with the higher CHROM confidence score and the later NTP timestamp. The 3 m merge radius was chosen based on the GPS CEP of the NEO-M8N (2.5 m, 50th percentile).

Isolated mode: if no message received for 5 consecutive seconds, the drone enters autonomous search mode and continues independently. On reconnection, full state synchronization is performed using the latest NTP timestamps as the tiebreaker.

5.5.3 Channel Models

Model	Parameters	Justification
Uniform Bernoulli	$p_{\text{loss}} = 0, 0.10, 0.30, 0.50, 0.70$	Standard baseline from prior UAV swarm literature
Gilbert-Elliott (burst)	Good state: $p_{\text{loss}} = 0.05$; Bad state: $p_{\text{loss}} = 0.80$; $P(\text{Good} \rightarrow \text{Bad}) = 0.10$; $P(\text{Bad} \rightarrow \text{Good}) = 0.25$; mean burst length = 4 packets	Empirically parameterized from 2.4 GHz WiFi measurements in a concrete-block test structure at 10 m range; approximates real disaster RF environment

VI. Execution: Construction and Testing

6.1 Simulation Experiments

6.1.1 Environment

Platform: ROS2 Humble + Gazebo Ignition on a lab desktop (Ubuntu 22.04, RTX 3060, 32 GB RAM). World: 200 m $\times$ 200 m with 10–20 survivors assigned random vitals from physiological distributions (HR: 45–140 bpm; RR: 8–30 brpm; temperature: 34.5–39.5°C), plus 3–5 static obstacle columns (simulating debris). UAV model: R.A.M.P.A.T. SDF file with mass, thrust, drag, and simple VTOL dynamics.

6.1.2 Simulation Trial Matrix

Factor	Levels	Trials/Cond.	Purpose
Protocol	Random, Centralized, Gossip (adaptive)	500	Primary comparison
Channel model	Bernoulli, Gilbert-Elliott	500 ea.	Bursty vs. i.i.d. loss
Packet loss	0%, 10%, 30%, 50%, 70%	500 ea.	Resilience curve
Survivor density	10, 15, 20 survivors	200 ea.	Scalability
Obstacle density	0, 3, 5 columns	200 ea.	Coverage realism

Total simulation runs: 3 protocols $\times$ 2 channel models $\times$ 5 loss levels $\times$ 3 densities $\times$ 3 obstacle levels $\times$ 200 trials = 10,800 runs (exceeds original 7,500; added obstacle and density factors address the ecological validity gap identified in pre-registration review). Statistical power at $n = 200$ per condition: > 0.99 for detecting a medium effect size ($d = 0.5$) with paired t-test.

6.1.3 Simulation Metrics

- Time to assess all survivors (primary outcome)
- Fraction of critical (Red) survivors found in first 25% of mission time
- Wasted scans (same survivor assessed twice within 60 s)
- Survivors missed (not assessed within mission time limit of 20 min)
- Total bytes exchanged (communication overhead)
- Task-allocation efficiency: (time under degraded comms) / (time under 0% loss), per protocol

6.2 Hardware Field Tests

6.2.1 Human Subjects Protocol

Volunteers: 10 healthy adults aged 18–45 (increased from 3–5 to provide adequate power for kappa analysis across all RATS score ranges). All volunteers provide written informed consent under the UCF IRB exemption. No identifiable data is retained; each volunteer is assigned a random ID. Testing is limited to vital signs measurable non-invasively at a distance.

To test the system in ranges that trigger Red and Yellow RATS categories (the clinically critical ranges), volunteers perform three standardized physiological maneuvers:

- Resting: volunteer lies still on mat. Expected HR 60–80 bpm, RR 12–18 brpm. RATS target: Green.
- Post-exertion: volunteer performs 90 seconds of step-ups immediately before assessment. Expected HR 100–140 bpm, RR 20–28 brpm. RATS target: Yellow to Red.
- Breath-hold (modified): volunteer breathes at controlled slow rate (6–8 brpm, metronome-guided) for 60 seconds. Expected RR 6–8 brpm. RATS target: Yellow.

This design ensures all three RATS color categories are represented in the dataset, allowing kappa to be computed in the ranges where triage decisions are most consequential. Volunteers are screened for cardiac or respiratory conditions; the physiological maneuvers are approved under the IRB exemption as minimal-risk exercise.

6.2.2 Ground Truth Instrumentation

Parameter	Ground Truth Device	Accuracy
Heart rate	Polar H10 chest strap (RR-interval)	± 1 bpm (validated vs. ECG in literature)
Respiratory rate	Nonin WristOx2 pulse oximeter (RR waveform output)	± 1 brpm in 6–60 brpm range

Skin temperature	Garmin Fenix 6 optical sensor + Fluke 62 MAX spot check	$\pm 0.5^{\circ}\text{C}$ (Fluke); $\pm 1^{\circ}\text{C}$ (Fenix 6)
Movement category	Independent human observer blind to drone score	Cohen's kappa between two observers pre-trial to confirm > 0.90

Note: The Garmin Fenix 6 replaces the previously proposed Garmin activity watch for respiratory rate, since the Nonin WristOx2 is a clinical-grade reference. The Fenix 6 is retained as the temperature reference; it is cross-checked against a Fluke 62 MAX IR thermometer spot measurement at the start of each trial to verify agreement within 1°C .

6.2.3 Field Test Conditions

ID	Configuration	Channel Model	Purpose
T1	1 drone, 5 vol., 0% loss	N/A	Baseline iPPG accuracy (all 3 algorithms compared; all maneuvers)
T2	2 drones, 5 vol., 0% loss	N/A	Two-drone consensus; basic coordination
T3	3 drones, 10 vol., 0% loss	Bernoulli	Full swarm; all 3 physiological conditions
T4	3 drones, 10 vol., 30% loss	Bernoulli	Moderate degradation
T5	3 drones, 10 vol., 50% loss	Bernoulli	Target degradation (primary hypothesis)
T6	3 drones, 10 vol., 50% loss	Gilbert-Elliott burst	Bursty channel; primary novelty condition
T7	3 drones, 10 vol., 0% loss (thermal only)	N/A	iPPG-disabled control; measures thermal-only triage accuracy

Each condition is repeated 5 times for a total of 35 field trials. With 10 volunteers per trial and 3 physiological conditions per volunteer, each trial yields 30 (volunteer, condition) assessment pairs. Total: $35 \times 30 = 1,050$ paired drone–ground truth comparisons for iPPG accuracy analysis. Statistical power for kappa > 0.80 at 1,050 pairs: > 0.99 for detecting kappa = 0.80 vs. null hypothesis of kappa = 0.60 (Fleiss, Levin, and Paik, 2003, sample size formula for kappa).

VII. Data Collection, Processing, and Statistical Analysis

7.1 Data Logged per Trial

- Timestamp (NTP-synchronized across all drones, max offset < 50 ms verified pre-trial)
- Drone ID, GPS position (lat/lon/alt), attitude (roll, pitch, yaw), battery voltage
- All survivor detections: GPS coordinates, thermal temperature (pre- and post-correction), CHROM HR, GREEN HR, POS HR, CHROM confidence, RR, movement classification, RATS score, assigned priority
- All inter-drone messages: sender ID, recipient ID, message size (bytes), send timestamp, ACK timestamp, packet loss flag
- Gimbal angles; ToF standoff distance; optical flow magnitude per frame
- FFC events: die temperature at trigger, correction offset applied
- Volunteer ID (anonymized), physiological condition (resting/post-exertion/breath-hold), ground-truth HR, RR, temperature, movement label, ground-truth RATS score

7.2 Statistical Analysis Plan

7.2.1 iPPG Accuracy

- Primary metric: MAE (bpm) for CHROM vs. Polar H10, computed separately for resting, post-exertion, and breath-hold conditions and at standoff distances of 5 m, 10 m, and 15 m.
- Bland-Altman plots for HR and RR: visualize bias and limits of agreement.
- CHROM vs. GREEN vs. POS: paired t-test on MAE; Bonferroni correction for 3 comparisons ($\alpha = 0.017$); effect size (Cohen's d).
- Hover-instability covariate: optical flow magnitude is included as a covariate in a mixed-effects model (volunteer as random effect) to quantify the marginal effect of drone motion on HR MAE.

7.2.2 Triage Agreement

- Cohen's weighted kappa (linear weights) between drone RATS score and ground-truth RATS score.
- Kappa computed separately for Green, Yellow, and Red subsets to confirm the system performs in clinically critical ranges.

- Confusion matrix: 3×3 (drone priority vs. ground truth priority); false-negative rate for Red survivors is the primary safety metric.

7.2.3 Swarm Performance

- Time-to-assess all survivors: Wilcoxon signed-rank test (non-parametric; distributions are right-skewed) comparing consensus vs. random baseline at each packet loss level.
- Task-allocation efficiency curves: plotted for Bernoulli and Gilbert-Elliott models separately; 95% confidence intervals from bootstrap resampling ($n = 10,000$ bootstrap samples).
- Gilbert-Elliott vs. Bernoulli comparison: paired t-test on task-allocation efficiency at matched mean packet loss levels.

7.2.4 Thermal Calibration Validation

- Post-correction temperature MAE vs. Fluke 62 MAX reference: computed per trial and pooled.
- Bland-Altman analysis of temperature bias as a function of die temperature (to confirm the correction model is adequate across the full flight temperature range).

VIII. Materials

8.1 Hardware per Drone (Three Drones Total)

- Custom STM32F405 PCB flight controller (already built for R.A.M.P.A.T.; two additional units fabricated)
- 3D-printed PLA/PETG frame, 4 × 2212 1000 kV motors, 4 × 30 A BLHeli ESCs
- FLIR Lepton 3.5 thermal camera module
- Arducam IMX519 RGB camera; 2-axis brushless gimbal
- VL53L1X ToF sensor; 4 × HC-SR04 ultrasonic sensors
- u-blox NEO-M8N GPS (PPS pin exposed for future chrony upgrade)
- MPU-6050 IMU; BMP388 barometer; 3 × 10 kΩ rotary potentiometers; INA219 current sensor
- NVIDIA Jetson Nano 4 GB; ESP8266 WiFi module; 915 MHz SiK telemetry radio; MicroSD
- 4S 5000 mAh LiPo battery per drone

8.2 Ground Station

- Laptop: Ubuntu 22.04, ROS2 Humble, React + D3.js dashboard, Python logging

- 915 MHz SiK telemetry radio (USB); RC transmitter (safety override); chrony NTP server

8.3 Validation Instruments (Human Subject Trials)

- Polar H10 chest strap × 10 (HR ground truth, ± 1 bpm)
- Nonin WristOx2 pulse oximeter × 10 (RR ground truth, ± 1 brpm)
- Garmin Fenix 6 watch × 10 (temperature ground truth)
- Fluke 62 MAX IR thermometer × 1 (pre-trial temperature cross-check)
- Metronome app (breath-rate pacing for breath-hold condition)

8.4 Software (All Free and Open-Source)

- ROS2 Humble (Jetson Nano + laptop); Gazebo Ignition (simulation)
- OpenCV 4.8, MediaPipe 0.10, SciPy 1.11, NumPy 1.25 (iPPG and thermal pipeline)
- TensorRT 8.6 (YOLOv8-nano and BlazePose on Jetson)
- chrony 4.4 (NTP clock synchronization)
- React 18 + D3.js 7 (ground station dashboard); Python 3.11 (data analysis, R 4.3 for statistical tests)

8.5 Estimated New Spending

Total new spending for two additional drones and all validation instruments: approximately \$1,200 (primary increases from original \$700 estimate: Nonin WristOx2 units × 10 at approximately \$40 each = \$400; 10 Polar H10 units already available via UCF lab loan). Cost per drone remains below \$1,000 (existing R.A.M.P.A.T. components reused; new units built at approximately \$850 each).

IX. Expected Outcomes and Novelty

9.1 Quantitative Expected Outcomes

- HR MAE (CHROM): 7–8 bpm at 10 m standoff, resting condition; 9–10 bpm post-exertion (degraded by motion artifact); remaining below the 10 bpm threshold in all conditions.
- RR MAE: 2.0–2.5 brpm at 10 m; within the 3 brpm threshold.
- Triage kappa: > 0.80 overall; > 0.75 in Red subset alone (harder due to small n at extreme physiology; reported separately).

- Swarm time-to-assess (0% loss): approximately 8 min for 10 survivors in 200×200 m field vs. 34 min random search (76% reduction; derived from simulation mean $\pm$ 95% CI; will be updated with field data).
- Task-allocation efficiency at 50% Bernoulli loss: > 85% of zero-loss baseline. At 50% Gilbert-Elliott burst loss: > 80% (anticipated slightly lower due to correlated burst; will be confirmed empirically).
- Movement classification kappa: > 0.85 vs. human observer.
- Thermal temperature MAE post-correction: < 0.8°C.

9.2 Novelty Contributions

- First demonstration of iPPG heart-rate measurement from a hovering VTOL drone at 5–15 m standoff.
- First UAV swarm validated under a Gilbert-Elliott bursty channel model (not only uniform random drop).
- First autonomous movement classification pipeline integrated into an aerial triage score.
- First experimental validation of the RATS score as a complete autonomous triage instrument.
- First multi-parameter aerial triage system at sub-\$1,000/drone cost with full field reparability.

9.3 Limitations and Future Work

- GPS-denied operation: the system requires GNSS. Future work will integrate visual odometry (ORB-SLAM3) for indoor or urban-canyon deployment.
- Healthy volunteer population: volunteers perform standardized physiological maneuvers to produce abnormal vital-sign ranges, but true trauma physiology (pain response, hemorrhagic shock) cannot be ethically simulated. The kappa claim must be understood as valid for the tested physiological range; clinical deployment would require further validation.
- RF model: the Gilbert-Elliott parameters are measured from a single concrete-block structure. Real disaster debris (rebar, steel) may produce worse burst behavior. Future work will measure RF propagation in additional structures.
- Single geographic site: all field tests are conducted at one UCF-authorized open field. Generalization to urban or forested terrain requires additional testing.

X. References

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